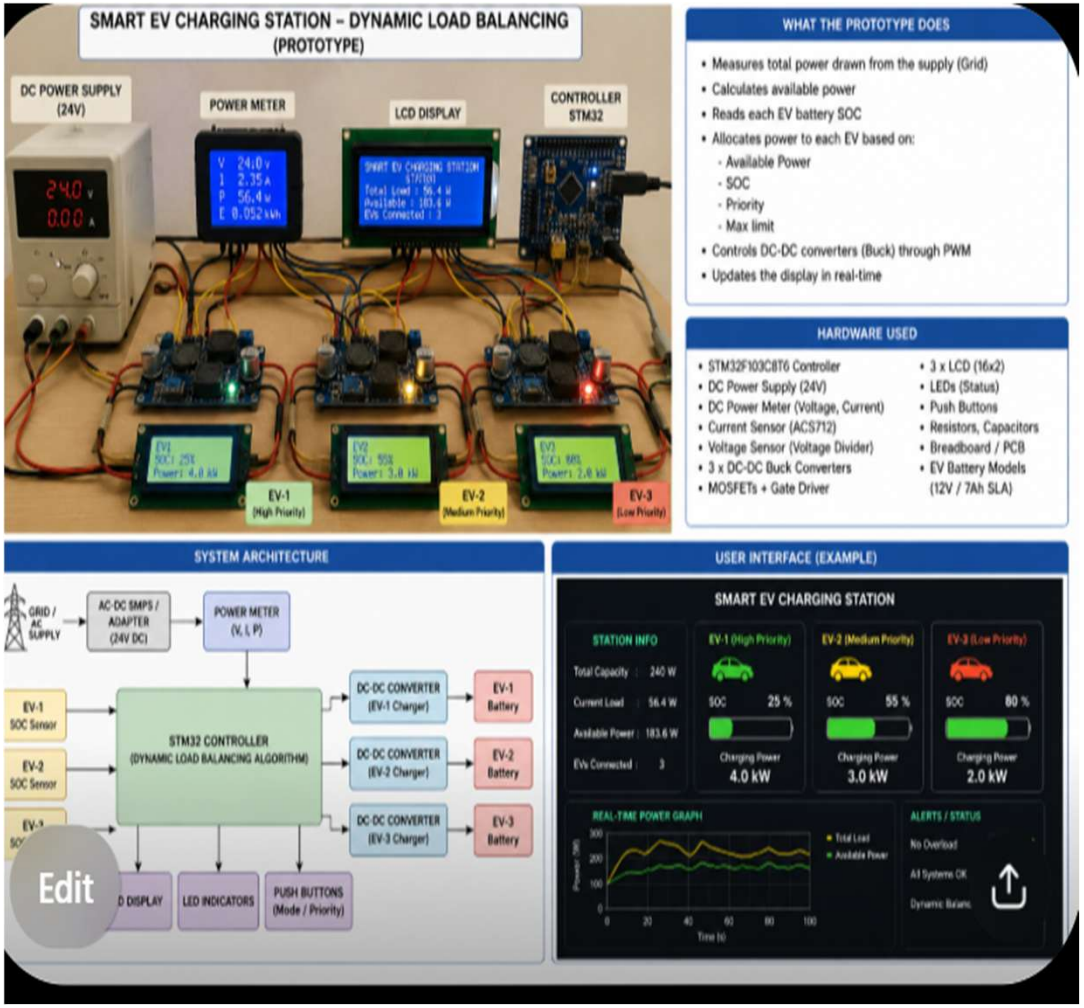
Innovative Design Project

SMART EV CHARGING STATION WITH DYNAMIC LOAD BALANCING

REAL WORLD PROBLEM

- EV charging stations experience uneven utilisation, long queues during peak periods and poor coordination with local grid capacity. The challenge is to build an intelligent platform that predicts charging demand, balances load across charging stations, recommends optimal charging schedules and reduces grid stress while improving user experience.
- Drivers frequently encounter occupied or non-functional charging stations. Develop a platform that provides real-time charger availability, reservation, navigation and estimated waiting times.



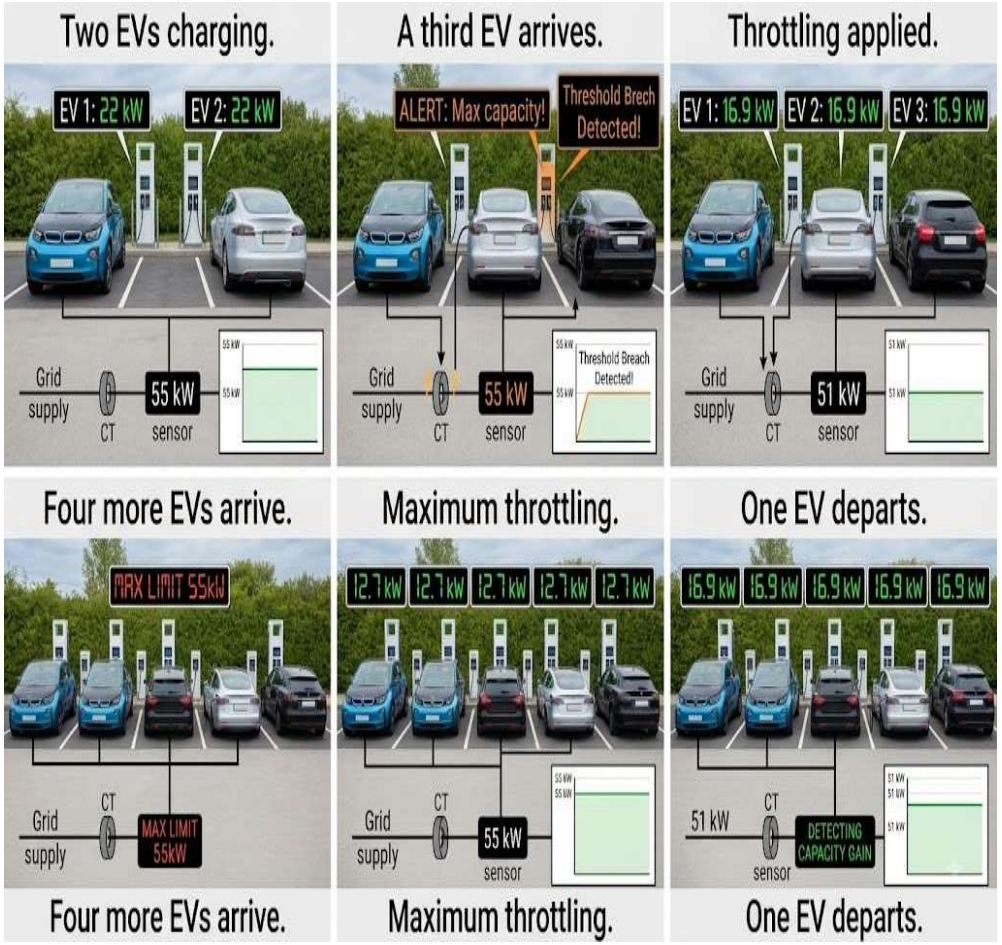
PROPOSED SOLUTIONS

1. Develop a smart EV charge station that continuously monitors the available electrical power and EV's charging demand , then dynamically distributes the available power among connected EV's

   Controller can consider

- The grid /load power availability
- No of connected EV's
- Maximum station capacity
- Peak load condition

2. SoC-Weighted Allocation: Implementation of an adaptive algorithm that assigns power proportional to battery depletion ($W_i = 100 - \text{SoC}_i$), prioritizing vehicles with critical state of charge (e.g., 10–20%) over nearly full vehicles (e.g., 70–90%)



NEED FOR PROPOSED SOLUTIONS

As electric vehicle (EV) adoption accelerates, power distribution infrastructure faces severe operational challenges. Standard charging setups operate on a "first-come, first-served" fixed-power model, creating massive strain on local electrical grids during peak demand hours.

1. Prevention of Local Grid Overloads & Transformer Trips

- An intelligent system that dynamically enforces a strict total grid power ceiling ($P_{\text{grid_max}} \leq 55$ kW) in real time to prevent circuit breaker trips, transformer burnout, and localized blackouts.

2. Fair & Optimized Power Allocation Based on Battery Urgency

- An adaptive State-of-Charge (SoC) Inverse Priority Algorithm ($W_i = 100 - \text{SoC}_i$) that prioritizes depleted batteries for emergency fast-charging while tapering power to nearly full vehicles, aligning with Li-ion charging physics.

SCOPE OF THE PROJECT

The scope defines the operational boundaries, technical inclusions, and practical limits of the proposed Smart EV Charging Infrastructure Optimization with Dynamic Load Balancing (DLB) and State-of-Charge (SoC) Priority Management.

A. Phase 1: Software Demonstration & Simulation Scope

Algorithmic Validation: Implementation and testing of the dynamic headroom tracking logic ($P_{\text{EV_Max}} = P_{\text{Grid_Limit}} - P_{\text{Building_Load}}$) to enforce a strict site capacity cap (e.g., 55 kW) in Python(using datafev library).

Inverse SoC Weighting Model: Simulation of the dynamic priority equation ($W_i = 100 - \text{SoC}_i$) across fluctuating vehicle arrival/departure sequences and varying initial battery states (10% to 90%).

Grid Impact Analysis: Verification of transformer load factors, thermal protection limits, and line voltage stability under balanced loads using ETAP or OpenDSS.

Interactive UI Makeup: A simulation dashboard displaying real-time power allocation graphs, active vehicle states, and capacity recovery events when vehicles disconnect.

B. Phase 2: Hardware Prototype & HMI Scope(Tentative)

Embedded Edge Controller: Porting and execution of the validated balancing control loop onto an ESP32 microcontroller for local, sub-second execution.

AC Grid Telemetry: Continuous current and active power monitoring of simulated site loads using non-invasive SCT-013 Current Transformers.

Vehicle State Input Simulation: Acquisition of simulated battery State-of-Charge (SoC) inputs via analog potentiometers / UI switches representing EV1–EVN.

Proportional Actuation & HMI: Signal generation for charging power regulation (IEC 61851 PWM duty cycles / multi-stage relays) alongside real-time system metrics displayed on an I2C LCD screen and an IoT dashboard (MQTT / Blynk).

WHAT WILL BE OUR EXPECTED OUTCOMES..

- Zero Grid Limit Breaches (Headroom Enforcement):

  The control algorithm guarantees that total charging station power draw never exceeds the preset grid ceiling ($P_{\text{Grid_Limit}}$, e.g., 55 kW), completely preventing circuit breaker trips and transformer overloads.

- Dynamic Low-Latency Power Redistribution:

  Power allocations dynamically update within milliseconds whenever baseline building loads fluctuate or when a new EV connects/disconnects.

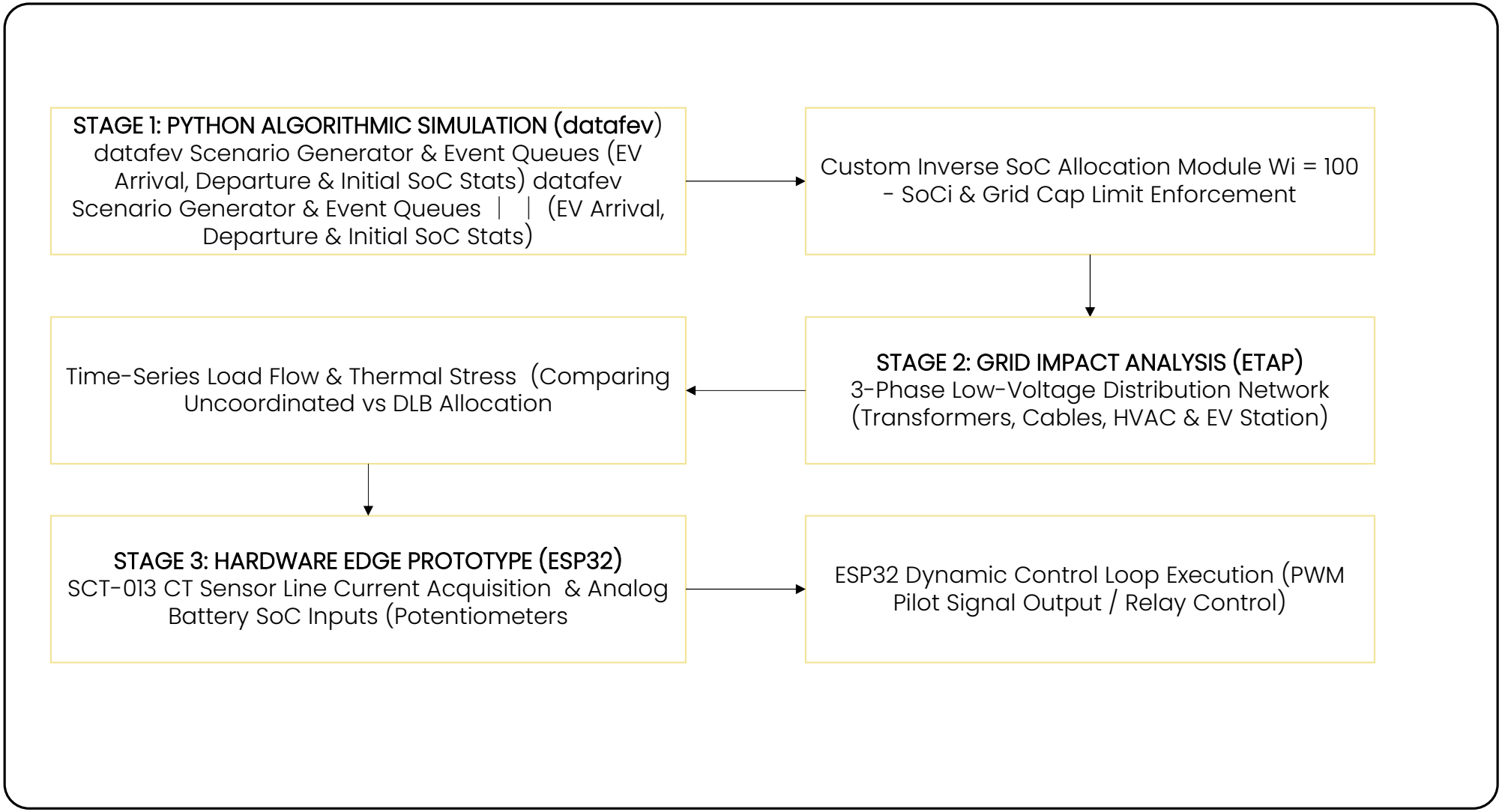
- Fair, Urgency-Based Allocation via Inverse SoC Weighting:

  Using $W_i = 100 - \text{SoC}_i$, vehicles with low battery states $(10\% - 30\%)$ are allocated maximum available charging power, while vehicles with high battery states $(70\% - 90\%)$ receive tapered trickle rates.

- Capacity Recovery:

  Whenever an EV finishes charging or unplugs, its freed capacity is instantly re-allocated to remaining depleted vehicles, maximizing overall station energy throughput.

PROPOSED METHODOLOGY


STAGE 1: PYTHON ALGORITHMIC SIMULATION (datafev)
datafev Scenario Generator & Event Queues (EV Arrival, Departure & Initial SoC Stats) datafev Scenario Generator & Event Queues | | (EV Arrival, Departure & Initial SoC Stats)
Custom Inverse SoC Allocation Module Wi = 100 - SoCi & Grid Cap Limit Enforcement
STAGE 2: GRID IMPACT ANALYSIS (ETAP)
3-Phase Low-Voltage Distribution Network (Transformers, Cables, HVAC & EV Station)
Time-Series Load Flow & Thermal Stress (Comparing Uncoordinated vs DLB Allocation
STAGE 3: HARDWARE EDGE PROTOTYPE (ESP32)
SCT-013 CT Sensor Line Current Acquisition & Analog Battery SoC Inputs (Potentiometers
ESP32 Dynamic Control Loop Execution (PWM Pilot Signal Output / Relay Control)